

Neural Networks and Deep Learning Lab 8

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1 Objective

To understand and implement a Generative Adversarial Network (GAN) to generate realistic images. Train a GAN on the MNIST dataset and analyze its performance based on generated samples and training loss curves.

2 Generated Images

GAN Generated Images after 50 Epochs



Figure 1: Images generated by the GAN after 50 epochs of training

The generated samples resemble handwritten digits from the MNIST dataset. Most of the digits are recognizable, although some images contain slight distortions or noise. This indicates that the generator has successfully learned the general structure of handwritten digits, but still has minor imperfections in some samples.

3 Training Curves

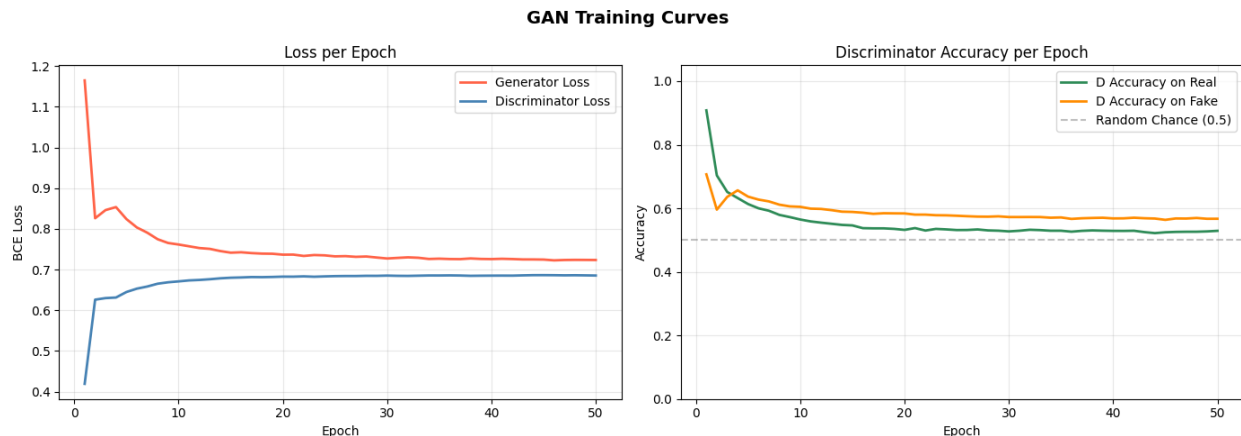


Figure 2: Generator and Discriminator training curves

The generator and discriminator losses stabilize as training progresses. The discriminator loss settles around approximately 0.52 while the generator loss stabilizes near 0.56. This indicates that both networks reach a relatively balanced adversarial state where neither model completely dominates the other.

4 Comparison of Real and Generated Images

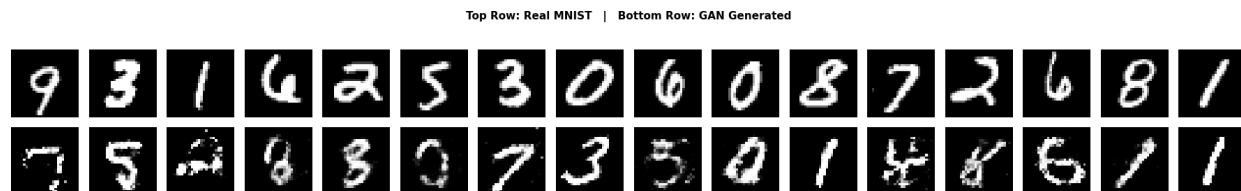


Figure 3: Top row: real MNIST images Bottom row: GAN generated images

Comparing the generated images with real MNIST samples shows that the generator has captured many important visual characteristics such as stroke thickness, digit shapes, and writing variations. However, some generated digits are slightly blurry or malformed.

5 Observations

- The generator gradually improves during training and begins producing recognizable digits after several epochs.
- The discriminator initially performs well at distinguishing real and fake images, but its performance decreases as the generator improves.

- The loss curves indicate a stable adversarial training process without major collapse.
- The generated images show diversity in digit shapes, suggesting that the model has learned multiple patterns present in the dataset.

6 Conclusion

The implemented GAN successfully learned the distribution of handwritten digits in the MNIST dataset. The generated images resemble real MNIST digits and demonstrate that adversarial training can be used to generate realistic synthetic data. Although some generated images contain minor distortions, the overall results show that the GAN is capable of producing visually plausible handwritten digits.